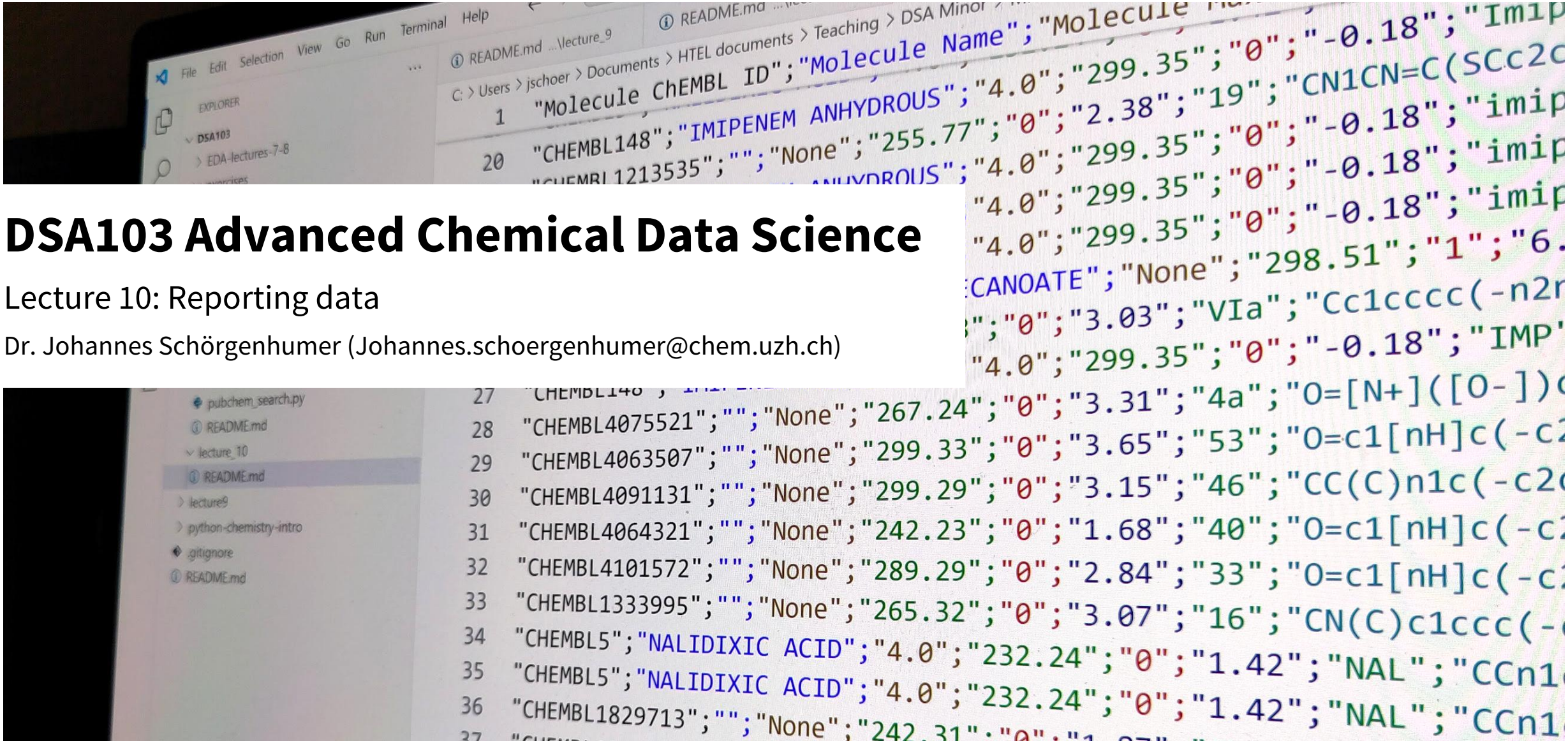
Universität Zürich[UZH]

Department of Chemistry



DSA103 Advanced Chemical Data Science

Lecture 10: Reporting data

Dr. Johannes Schörgenhumer (Johannes.schoergenhumer@chem.uzh.ch)

Content overview

FAIR principles

Practical aspects of data reporting

Ethical and legal considerations (CARE)

Challenges and outlook

Motivation

- Data reporting ensures transparency, reproducibility, and reuse
- Poorly reported data = wasted effort, unreliable science, no reuse
- Data = a product others need to understand and trust
- Good reporting = essence of research integrity

The FAIR principles

Findable
- Global unique identifier (e.g. DOI)
- Data described by rich metadata
- Metadata include identifier
- (Meta)data registered or indexed in searchable resource
- Machine-readability!

Accessible
- Data retrievable by identifier
- Retrieval protocol open with authorisation and authentication where possible
- Open vs. Controlled access

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Reusable
- License (CC-BY, Open, …)
- Detailed provenance
- Domain-relevant community standards
- Detailed context (Readme)

Interoperable
- Use of accessible, open standards
- Standard vocabularies and formats (JSON, CSV, XML, …)
- Language

https://www.go-fair.org/fair-principles/

Short Activity / Discussion: Convincing Sceptics

Each person/group will get a FAIR principle to work on.

Find convincing pro-arguments for FAIR data
Think about practical challenges and drawbacks that you see with the other three FAIR principles.

Present your FAIR aspect to the others who are to challenge your FAIR principle! Can you convince the sceptics in this short discussion with the arguments you prepared?

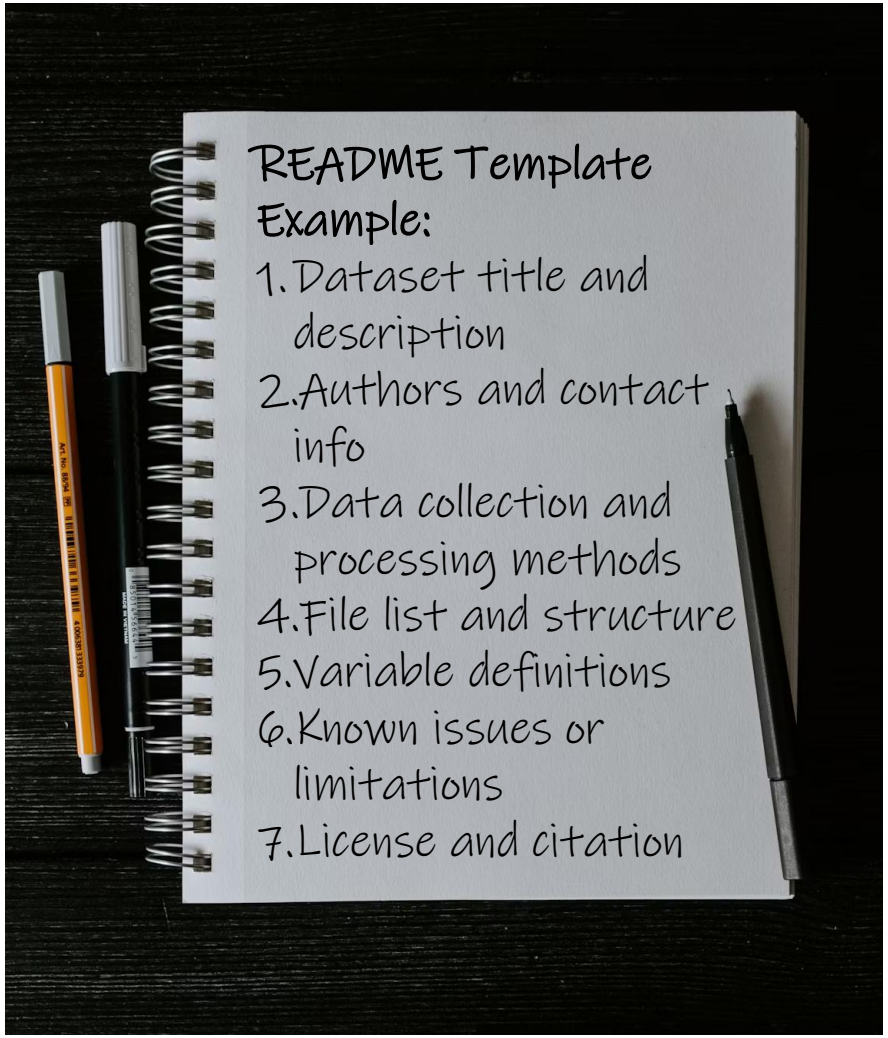
Some benefits:

- Increased reproducibility
- Easier collaboration across labs/disciplines
- Higher visibility & citations
- Facilitates machine learning & AI applications

Practical aspects of data reporting – documentation:

Data documentation
(metadata, lab notebooks, README files)

- Metadata = data about data; descriptive (what), structural (how), administrative (who/when)
- Readme files: Explain files, variables, units, and known issues
- Data dictionaries/codebooks: Define variables, categories, units
- Methods & context: Include how, when, and where data were collected



Practical aspects of data reporting – data standards & formats:

Use open, non-proprietary formats:

- Text: .csv, .tsv, .json
- Images: .tiff, .png
- Structured data: e.g. .xml, .netCDF

Discipline-specific standards:

- Chemistry: e.g. ORD
- Genomics: FASTA
- …

Controlled vocabularies to improve interoperability

Practical aspects of data reporting – versioning & provenance:

To track changes and reproduce results

- Version control systems: e.g. Git
- Semantic versioning: v1.0 → v1.1 → v2.0
- Changelogs: Record what was updated and why (commit messages!)
- Provenance tracking: Where data came from and how it was transformed

Practical aspects of data reporting – repositories & citability:

Deposit final datasets in trusted repositories:

- General repositories, e.g. Zenodo, Figshare
- Domain-specific repositories, e.g. Medline, PDB, SDBS
- Institutional repositories, e.g. Research Collection (ETH)

Citability:

- Obtain a DOI for citation
- Link data, code, and publication
- Include version in the citation (important for reproducibility)

Practical aspects of data reporting – quality control:

Goal: Ensure accuracy, consistency, completeness.

- During collection: e.g. instrument calibration, double-entry checks
- During processing: e.g. validation rules, range checks, outlier detection
- During analysis: e.g. reproducible pipelines, cross-validation
- Document all QC steps transparently

Stage	QC Activity	Example
Collection	Double data entry	Manual survey entry check
Processing	Range check	Flag all temperatures > 60 °C
Analysis	Re-run pipeline	Verify output on different system

Practical aspects of data reporting – ethical & legal considerations:

- Data privacy
- Consent for data sharing
- **CARE** – Principles for Indigenous Data Governance



- Licensing & intellectual property (e.g. Creative Commons, open vs. restricted)
  - BY: credit must be given to the creator.
  - SA: Adaptations must be shared under the same terms.
  - NC: Only noncommercial uses of the work are permitted.
  - ND: No derivatives or adaptations of the work are permitted.
  - CC0: Public domain dedication

https://creativecommons.org/share-your-work/cclicenses/
https://www.gida-global.org/care

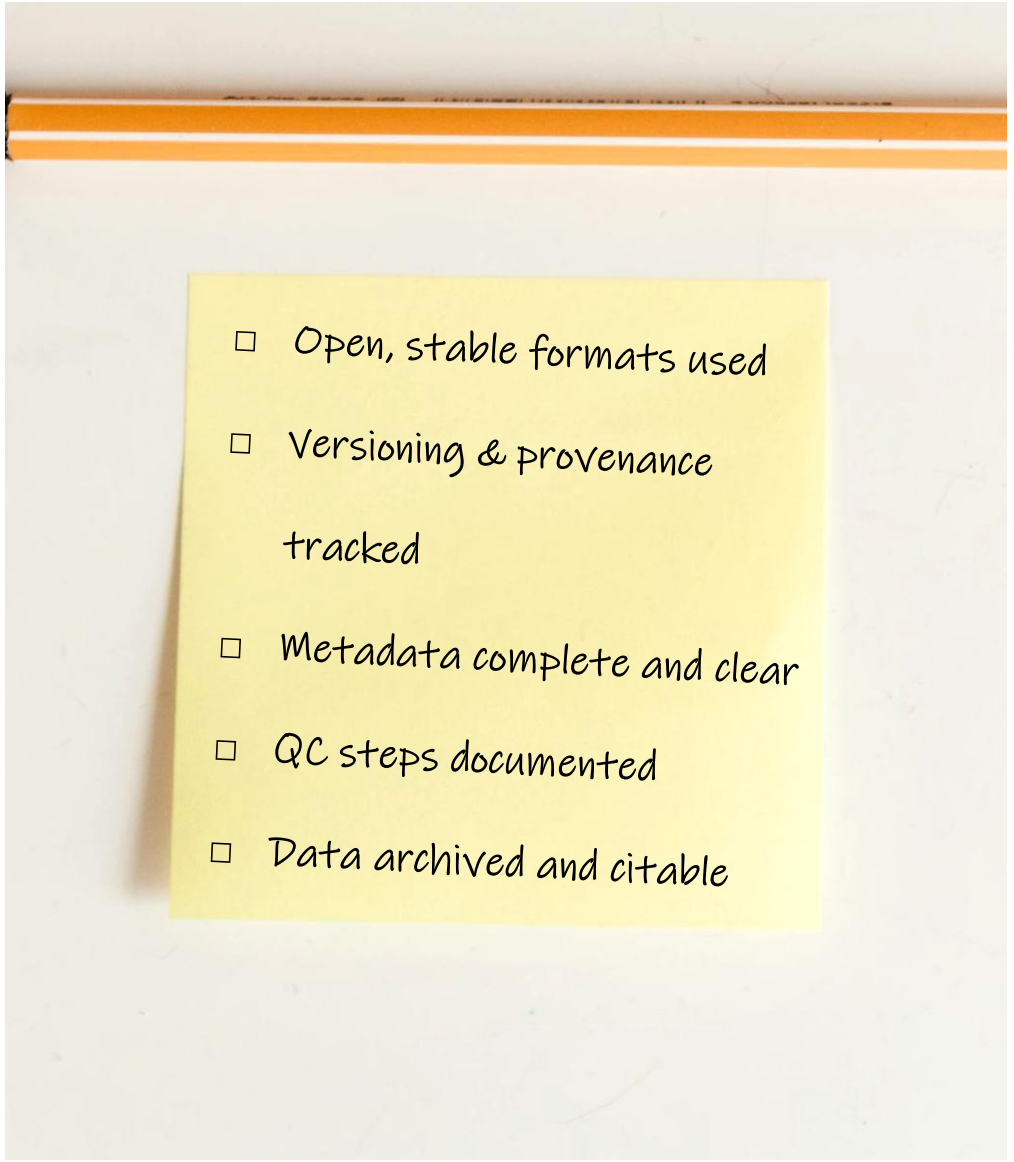
Challenges & outlook:

Common barriers:

- Time, Cost
- Lack of training
- (Cultural) resistance (IP, …)
- Ambiguity (“FAIR-washing”)
- Burocracy

Some solutions:

- Training, institutional support, automation tools, community standards.
- Try to establish and maintain high personal working standard
- Good data reporting = FAIR + transparent + trustworthy



https://www.ub.uzh.ch/en/wissenschaftlich-arbeiten/mit-daten-arbeiten/daten-ablegen-und-teilen.html

Acitivity: Assessing FAIR Principles

Consider some examples:

https://www.science.org/doi/10.1126/science.aap9112
https://github.com/Paulilein/Perera2018
https://pubs.rsc.org/ta/content/articlehtml/2024/md/d4md00196f
https://zenodo.org/records/17370136
https://zenodo.org/records/17369268
https://zenodo.org/records/11611178

Evaluate against FAIR principles:

- Is it Findable? (Does it have a DOI?)
- Accessible? (Open license, protocols?)
- Interoperable? (Standard formats?)
- Reusable? (Metadata, license, documentation?)